Coral Reef Regime Classification

Problem statement

Coral reefs are vital marine ecosystems but face growing stress from human and natural influences (warming, turbidity, storms). This project predicts coral reef regimes — categorical reef health states (Low, Medium, High) — from environmental and reef survey variables to support monitoring and management decisions.

Approach

I framed the task as a supervised multiclass classification problem. From a large compiled reef table (~41k records), I selected numeric predictors (depth, percent cover, bleaching metrics, sea-surface temperature indices, winds, etc.), handled missing values with mean imputation, standardized features, and addressed class imbalance using SMOTE. Models evaluated included:

- Logistic Regression (baseline, RandomizedSearchCV, GridSearchCV)
- Support Vector Machine (baseline, HalvingGridSearchCV, PCA+SVM)
- Random Forest (baseline)

Implementation overview

Data cleaning and preprocessing were executed in dedicated notebooks. Key steps:

- Load CSV, replace non-data tokens ('nd') with NaN, coerce numeric types.
- Drop rows missing core targets (Percent_Bleaching / Percent_Cover); impute remaining NaNs withcolumn means.
- Create multiclass target ('Low','Medium','High') via Percent_Bleaching bins.
- Train/test split (80/20, random_state=42, stratified by class).
- Scale numeric features with StandardScaler; apply SMOTE to training fold.

Modeling:

- Logistic Regression: baseline; tuned with RandomizedSearchCV (fast sampling) and GridSearchCV(exhaustive) — RandomizedSearchCV produced the best LR (~66%).
- SVM: baseline RBF, tuned with HalvingGridSearchCV (fast successive-halving), and PCA+SVM (95%variance). HalvingGridSearchCV gave the best SVM (~94%); PCA+SVM improved runtime and achieved ~85%.
- Random Forest: baseline ensemble (n_estimators=200, class_weight='balanced') — best performer at 98.13% accuracy.

Key results (ranked by test accuracy)

1. Random Forest (baseline) — 98.13%
2. SVM (HalvingGridSearchCV) — 94%
3. SVM (PCA+SVM) — 85%
4. SVM (baseline) — 80%
5. Logistic Regression (RandomizedSearchCV) — 66%
6. Logistic Regression (GridSearchCV) — 54%
7. Logistic Regression (baseline) — 53%

Conclusions

The Random Forest baseline showed the strongest predictive performance on the assembled dataset, suggesting the target patterns are well-captured by an ensemble of decision trees without heavy tuning. SVM with HalvingGridSearchCV provided a close second, proving that focused hyperparameter search can yield near-ensemble performance. Logistic Regression — even after tuning — underperformed relative to non-linear methods, indicating non-linear relationships and interactions are important for this dataset.

Challenges

- Missing and mixed-type fields required careful coercion and imputation; many rows contained 'nd' placeholders.
- Class imbalance necessitated SMOTE; evaluation used stratified splits and weighted metrics.
- Hyperparameter searches can be computationally expensive;